the electron is spinning around its own axis, in addition to its orbital motion around the nucleus. Due to these motions, a tiny magnetic field is generated and it is measured in terms of magnetic moment. On the basis of magnetic properties, materials can be broadly classified as (i) paramagnetic materials (ii) diamagnetic materials, besides these there are ferromagnetic and antiferromagnetic materials.

Materials with no elementary magnetic dipoles are diamagnetic, in other words a species with all paired electrons exhibits diamagnetism. This kind of materials are repelled by the magnetic field because the presence of external magnetic field, a magnetic induction is introduced to the material which generates weak magnetic field that oppose the applied field.

Paramagnetic solids having unpaired electrons possess magnetic dipoles which are isolated from one another. In the absence of external magnetic field, the dipoles are arranged at random and hence the solid shows no net magnetism. But in the presence of magnetic field, the dipoles are aligned parallel to the direction of the applied field and therefore, they are attracted by an external magnetic field.

Ferromagnetic materials have domain structure and in each domain the magnetic dipoles are arranged. But the spin dipoles of the adjacent domains are randomly oriented. Some transition elements or ions with unpaired d electrons show ferromagnetism.

3d transition metal ions in paramagnetic solids often have a magnetic dipole moments corresponding to the electron spin contribution only. The orbital moment L is said to be quenched. So the magnetic moment of the ion is given by

$$\mu = g\sqrt{S(S+1)}\ \ \mu_B$$

Where S is the total spin quantum number of the unpaired electrons and is μ_B Bohr Magneton.

For an ion with n unpaired electrons $S = \frac{n}{2}$ and for an electron g=2

Therefore the spin only magnetic moment is given by

$$\mu = 2\sqrt{\left(\frac{n}{2}\right)\left(\frac{n}{2}+1\right)}\ \ \mu_B$$

$$\mu = 2\sqrt{\left(\frac{n(n+2)}{4}\right)}\ \ \mu_B$$

$$\mu = \sqrt{n(n+2)}\ \ \mu_B$$

The magnetic moment calculated using the above equation is compared with the experimental values in the following table. In most of the cases, the agreement is good.

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